

8D PROBLEM-SOLVING REPORT

AUTOMOTIVE TIER-1 QUALITY ENGINEERING & ROOT CAUSE ANALYSIS

REPORT REF:

8D-ECB-2026-01

LEAD ENGINEER:

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Company Context: Automotive Tier-1 Casting & Precision Machining (Bosch, Bharat Forge, Sundram Fasteners, Endress+Hauser, etc.)

Component Details: Engine Cylinder Block / Brake Disc

Target Specification: Surface Roughness $R_a \leq 1.6 \mu m$ | **Observed Defect:** Surface Finish Failure ($R_a = 3.2 \mu m - 4.5 \mu m$) |

Rejection Rate Spike: 1.2% → 6.8%

D1: ESTABLISH TEAM (CROSS-FUNCTIONAL TEAM)

Formed an inter-departmental Cross-Functional Team (CFT) leveraging Agile methodology and departmental balance:

1. **Automobile Department**
2. **Operations Department**
3. **Core Management Department** (For proper implementation of Agile methodology & cross-functional team balance)
4. **Manufacturing Department**
5. **Industrial Department**
6. **Supply Chain Department** (For material check with respect to parts)

Toyota Philosophy Integration:

If we take suggestions from vendors for good material, it will also help to find the best material for automotive parts. Involving material suppliers directly ensures defect prevention at the raw material stage.

7. **Quality Inspection / QA / QC Department**
8. **Production Department**

D2: PROBLEM DEFINITION & DESCRIPTION

Problem Statement: In an Automotive Tier-1 casting and precision machining setup (Bosch, Bharat Forge, Sundram Fasteners, Endress+Hauser), facing a major problem of surface finish with respect to Automotive Brake Disc / Cylinder Block face which exceeds more than $1.6 \mu m$ than its actual limit, which causes rejection rate increment from 1.2% to 6.8%, preventing customer retention as well as maximizing customer escalation.

Describing Problem (5W2H Framework):

1. **What:** Surface finish standard target $R_a \leq 1.6 \mu m$ (Ideal / Standard). Actual measured finish: $R_a = 3.2 \mu m$ to $4.5 \mu m$.

Taguchi Quality Loss Philosophy:

According to Taguchi's Quality Loss Function ("Smaller-is-Better" characteristic), driving surface roughness R_a as far below the upper specification limit of $1.6 \mu m$ as possible minimizes variation and prevents customer quality loss.

2. **Where:** Detected during casting inspection & at manufacturing line / production point.
3. **When:** When the parts were getting rejected at the final stage inspection.
4. **Who:**
 - **1st Level:** Producer (Who is responsible + Caster for matching parts) + Quality Inspection (QI).
 - **2nd Level:** Customer escalations.
5. **Why:**
 - Surface finish exceeds its actual limit due to defects: porosity, holes, gaps, or micro-cracks.
 - Caused if the material inside the parts is not watertight (detected during CAD modelling).
 - Caused if the material does not settle easily during parts making.
6. **How Many:** Rejection rate increased from 1.2% → 6.8%, which causes customer escalation due to vibration, noise, etc.

D3: INTERIM CONTAINMENT ACTIONS (ICA)

1. Stop the production immediately at the manufacturing line.
2. Isolate all affected non-conforming parts.
3. Kept the parts under Red Zone and assigned Red Flagging / Tagging [ISO 9001:2015 Clause 8.7](#).

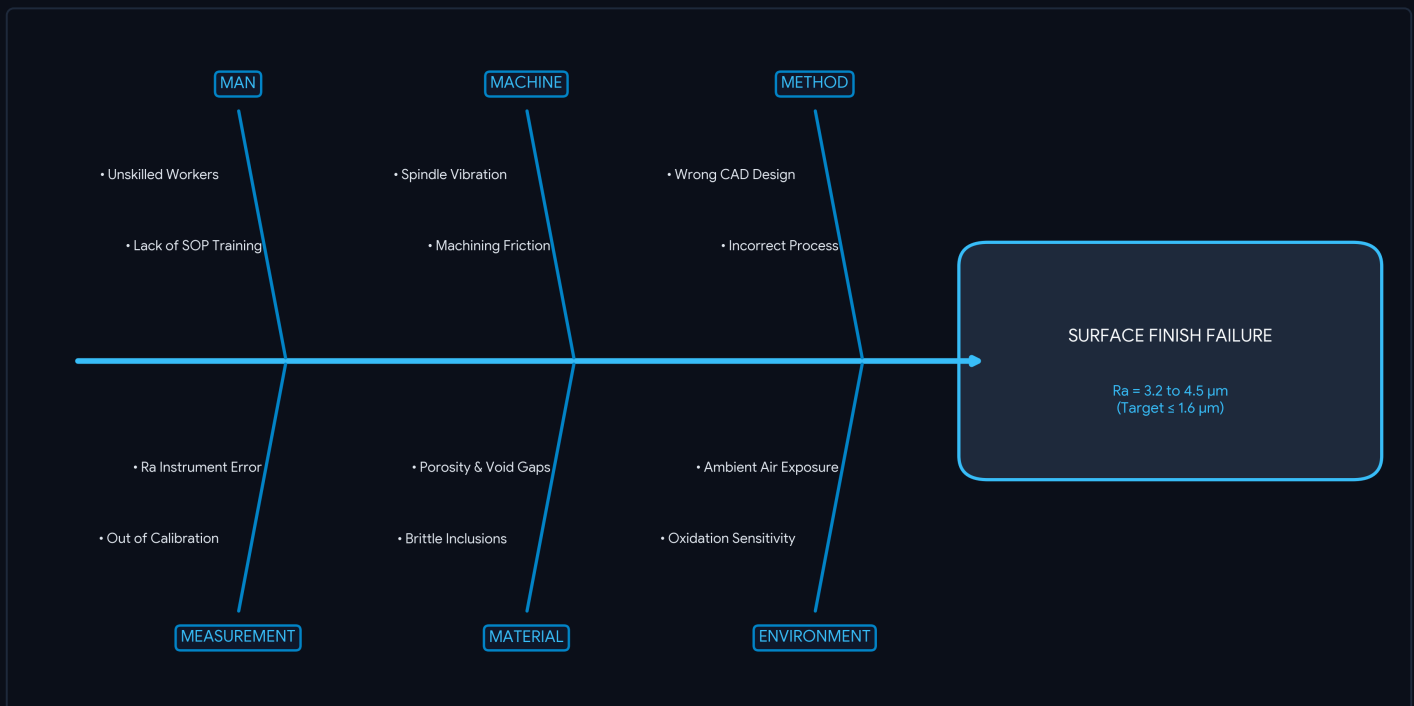
D4: DETERMINE & VERIFY ROOT CAUSE (RCA)

Applied multi-tool Root Cause Analysis using **Fishbone (Ishikawa) Diagram + Pareto Chart + FMEA Analysis**.

1. Fishbone (Ishikawa) 6M Breakdown:

- **(A) Man:** Due to unskilled / unspecialized workers (Lack of training).
- **(B) Measurement:** Not proper calibration or lack of re-enhancement of R_a measuring instrument (Profilometer).
- **(C) Machine:** Spindle vibration, noise, friction during machining.
- **(D) Material:** Rough surface material, brittle or hard material causes imbalance in surface finish (Porosity, irregular shape, size).
- **(E) Environment:** If the material with which parts/components are made is sensitive to oxidation, it will cause poor surface finish when exposed to air.
- **(F) Method:** Wrong modelling design of brake component / wrong selection of method for design process.

Fishbone Visual Breakdown Diagram:



2. Pareto Chart (80/20 Rule) Prioritization:

Measurement > Method > Machine > Material > Man > Environment

3. FMEA Analysis (Failure Mode and Effects Analysis):

- **Layers:** System → Overall on 6M | Design → Machine | Process → Method + Measurement
- **RPN Formula:** $RPN = Severity (S) \times Occurrence (O) \times Detection (D)$
- **Purpose:** Used for finding maximum failure points and prioritizing corrective measures.

D5: CHOOSE & VERIFY PERMANENT CORRECTIVE ACTIONS (PCA)

1. Use 5S Methodology + Kaizen (Toyota Benchmarking):

- Sort, Set in order, Shine, Sustain, Standardize + Kaizen (Continuous Improvement).
- 5S is used for preventing failure at maximum points obtained from FMEA analysis.

2. **Apply Six Sigma DMAIC:** Applied for controlling oxidation, porosity, and variability.

D6: IMPLEMENT & VALIDATE PCA

1. **PDCA Cycle:** Use PDCA for implementation.
2. **System Standardizing:** For standardizing, use auditing of system / process.
3. **SPC Control Charts:** Use of control charts — *X-bar Chart*, *S Chart*, and *R Chart* to monitor continuous variable surface roughness data (R_a).
4. **Data Implementation & Advanced NDT:** Use check sheet for implementation of data got from control chart. For preventing surface roughness increment, use 3D printing & **Micro X-Ray (Industrial CT Scanning)** to non-destructively detect internal porosity, holes, and void gaps early.

D7: PREVENT RECURRENCE (SYSTEMIC PREVENTION)

1. **Prepare SOPs:** Prepare SOPs based on SPC data (Reformation of SOPs based on overall analysis, customer feedback, and specification limit).
2. **ISO 19011 Compliance:** Follow auditing principle standard given under **ISO 19011**.
3. **Benchmarking:** Benchmarking with top companies in Function / Machine / Method / System.
4. **Zero-Defect Philosophy & Juran's Trilogy:** Follow Zero-Defect System Philosophy of Philip Crosby together with implementing Juran's Trilogy (Quality Planning, Quality Control, Quality Improvement).
5. **Poka-Yoke:** Integrate Poka-Yoke (Mistake-proofing) on the shop floor.

D8: RECOGNIZE TEAM & SYSTEM CLOSURE

1. Official signature of all Department Heads for formal closure.
2. Recognize, empower, give authority, and appreciate team members for their outstanding approach & hard work.
3. Follow Dr. W. Edwards Deming management philosophy for long-term continuous transformation.

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